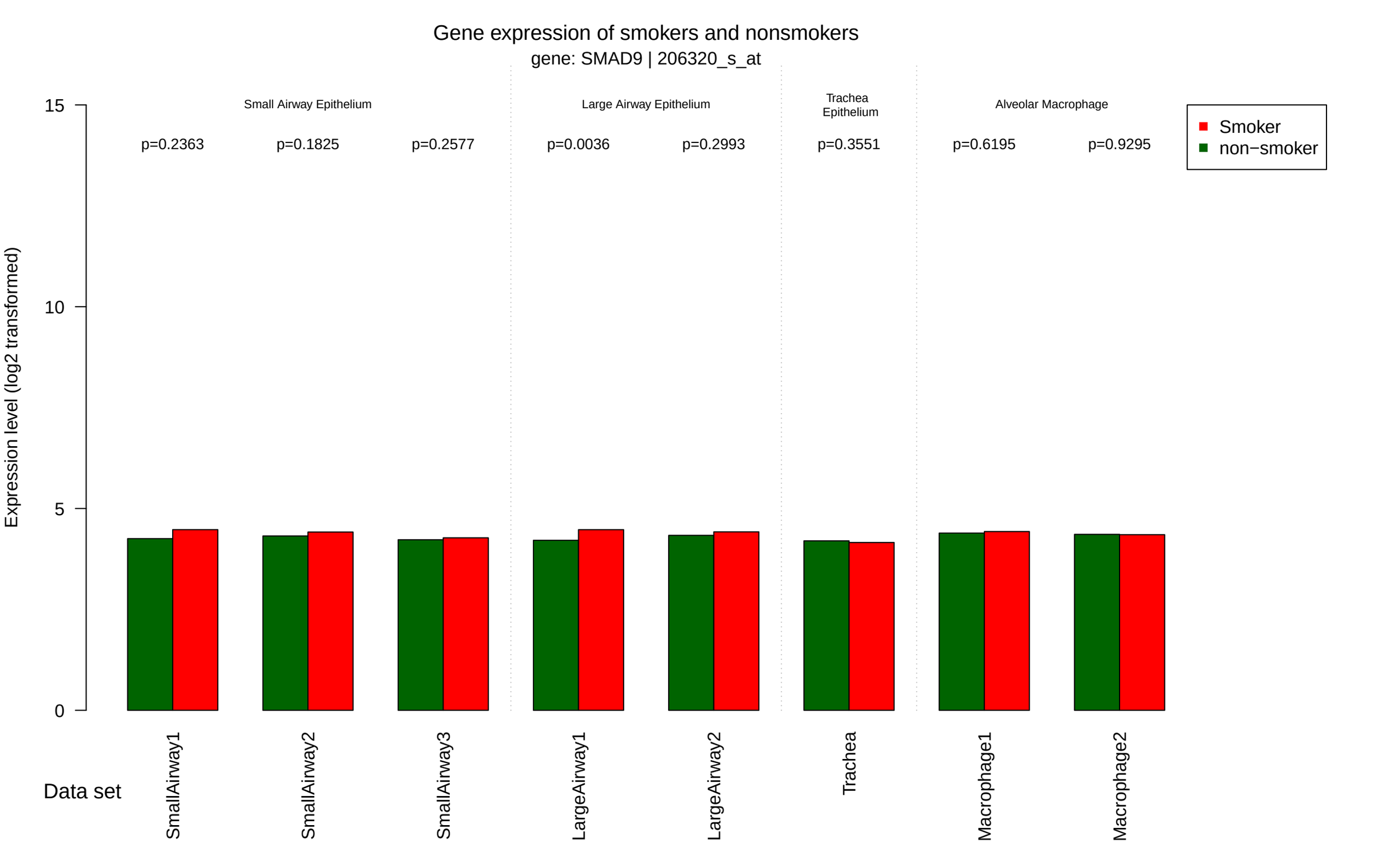
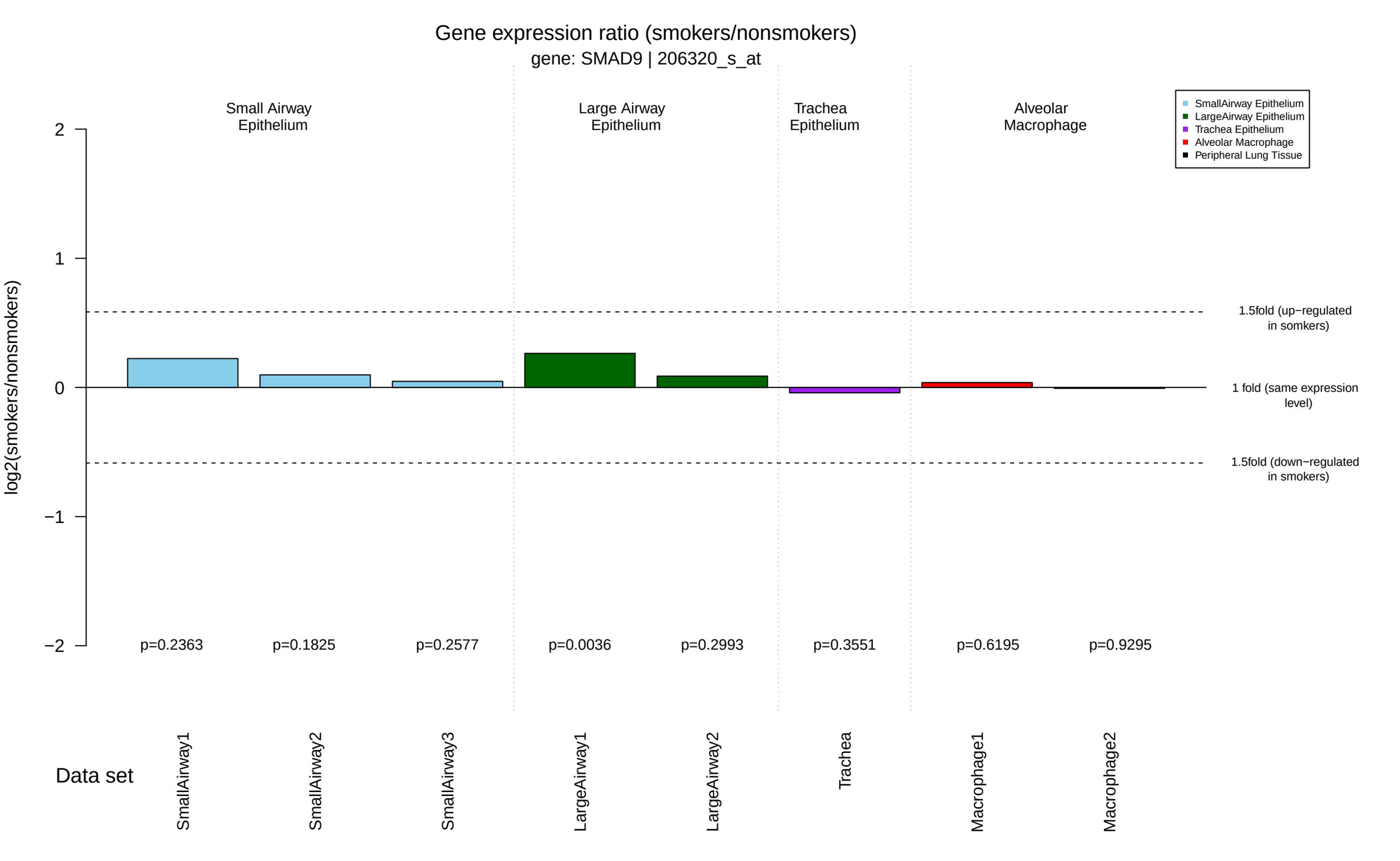
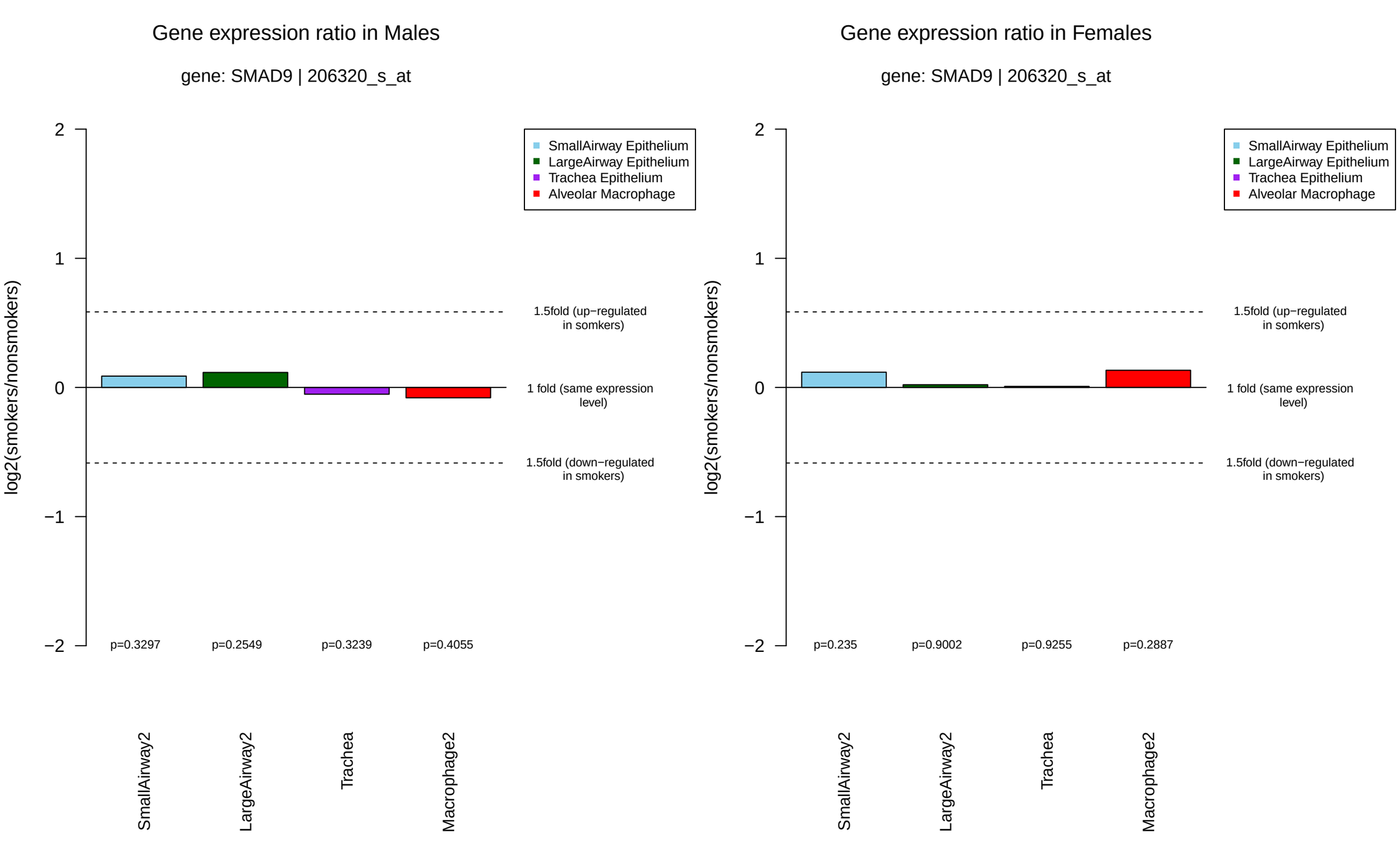
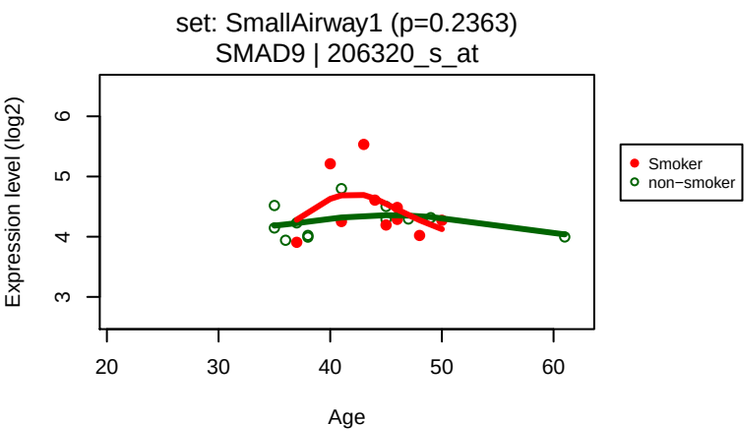
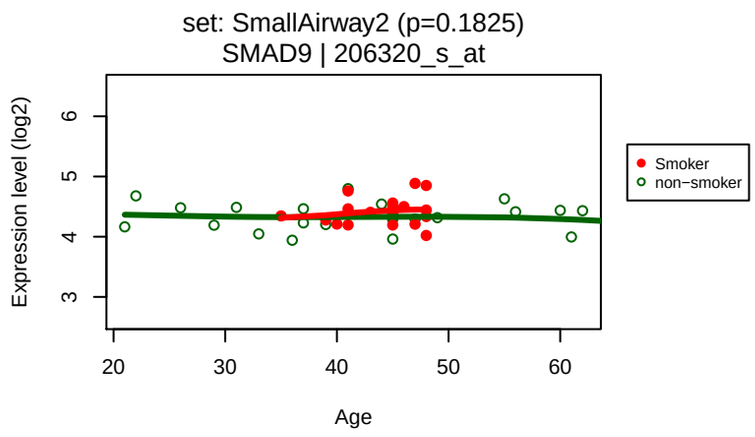
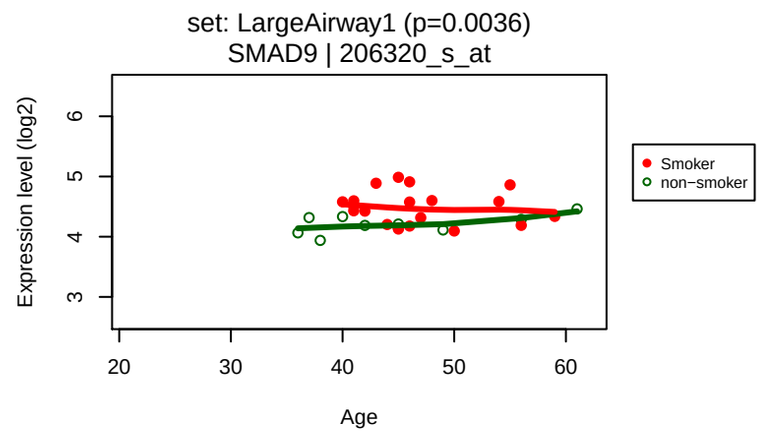

Gene expression of smokers and nonsmokers
gene: SMAD9 | 206320_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2363
p=0.1825
p=0.2577
p=0.0036
p=0.2993
p=0.3551
p=0.6195
p=0.9295
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

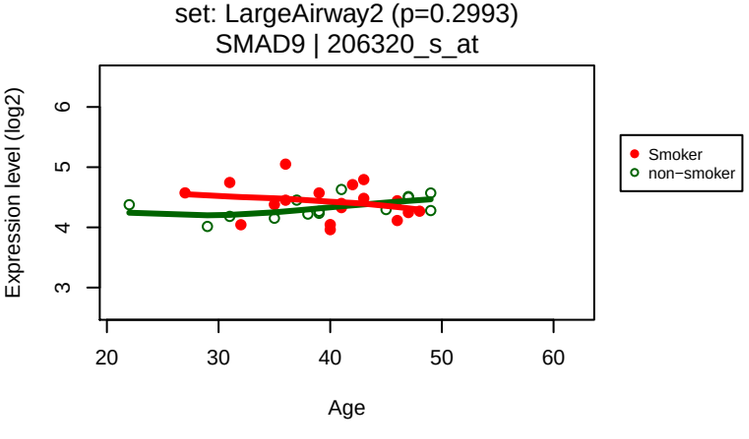

Gene expression ratio (smokers/nonsmokers)
gene: SMAD9 | 206320_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2363
p=0.1825
p=0.2577
p=0.0036
p=0.2993
p=0.3551
p=0.6195
p=0.9295
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

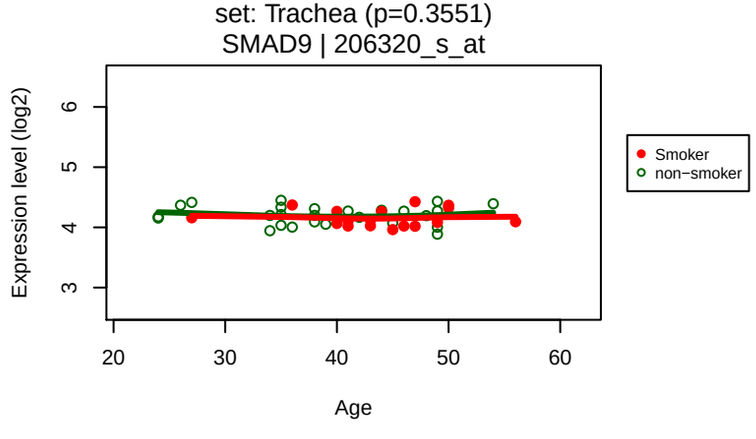

Gene expression ratio in Males
gene: SMAD9 | 206320_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3297
p=0.2549
p=0.3239
p=0.4055
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SMAD9 | 206320_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.235
p=0.9002
p=0.9255
p=0.2887
SmallAirway2
LargeAirway2
Trachea
Macrophage2

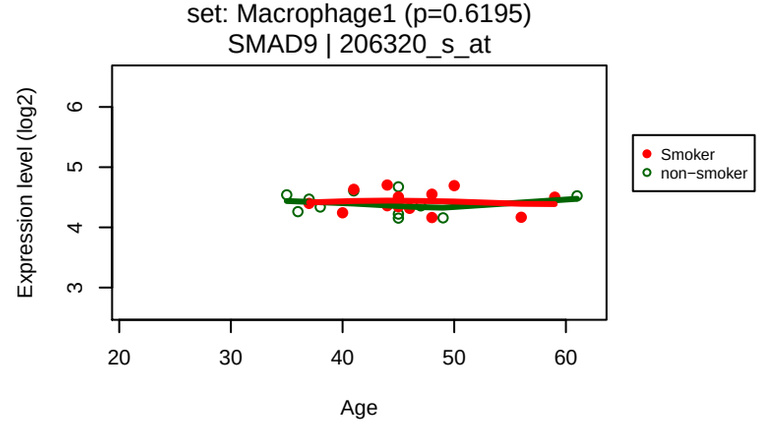

set: SmallAirway1 (p=0.2363)
SMAD9 | 206320_s_at
Expression level (log2)
Age
Smoker
non−smoker

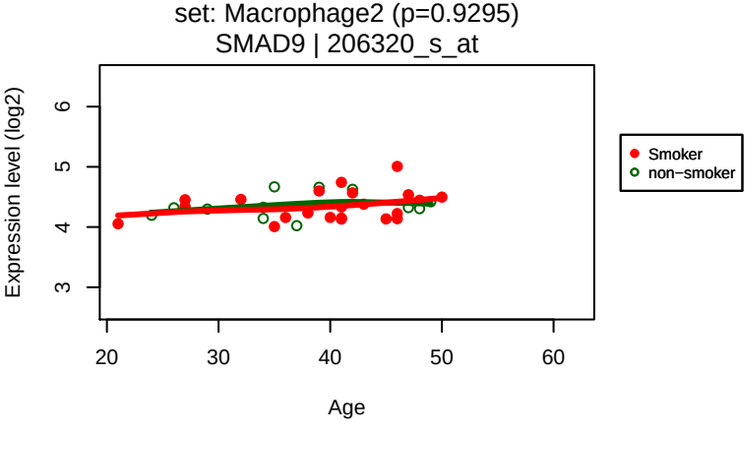

set: SmallAirway2 (p=0.1825)
SMAD9 | 206320_s_at
Expression level (log2)
Age
Smoker
non−smoker

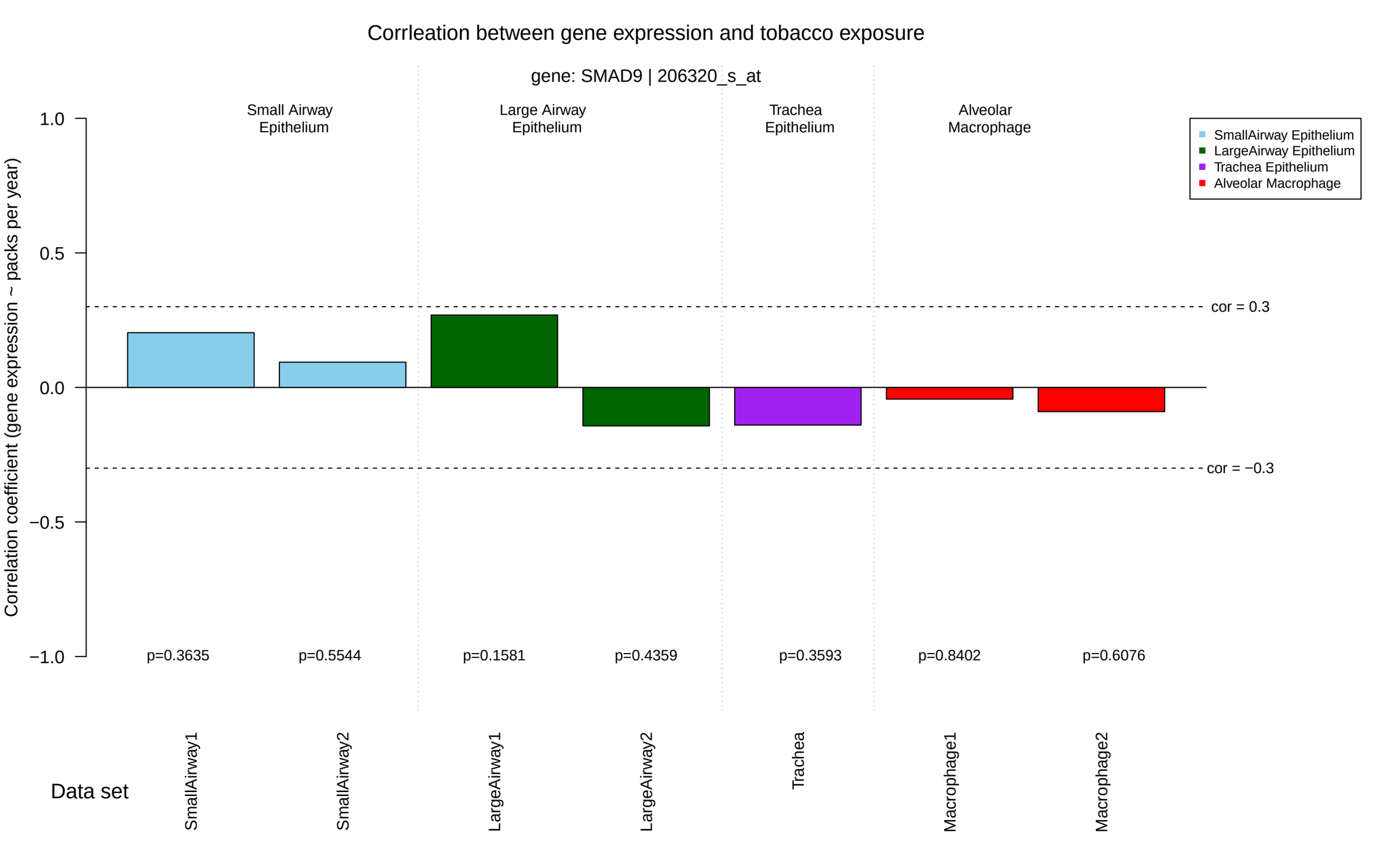

set: LargeAirway1 (p=0.0036)
SMAD9 | 206320_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2993)
SMAD9 | 206320_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3551)
SMAD9 | 206320_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6195)
SMAD9 | 206320_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9295)
SMAD9 | 206320_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SMAD9 | 206320_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3635
p=0.5544
p=0.1581
p=0.4359
p=0.3593
p=0.8402
p=0.6076
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2